

No Detectable Degradation of Dense Multilingual Retrieval with Language Resource Level: A Pre-Registered Measurement Study Across 18 Languages

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ABSTRACT

Multilingual dense retrieval is widely reported to degrade for low-resource languages, with tokenizer over-segmentation frequently offered as the explanation. We test both claims in a pre-registered study across all 18 languages of MIRACL, equalizing collection size at 49,043 passages per language (MIRACL’s native corpora span a 271-times size ratio) and capping evaluation queries so judged-document proportion does not itself vary with language. Against a uniformly applied BM25 reference, a retrieval-trained multilingual encoder is superior in every language it declares support for: a pooled advantage of 0.2627 nDCG at 10 (95 percent confidence interval 0.2240 to 0.3013, 17 languages; 0.2464 excluding Chinese, whose lexical reference fails on 51 percent of queries). Across roughly 76,000 times in pretraining corpus volume we find no sign of the expected degradation: the three lowest-resource languages show advantages of 0.2096 to 0.2873, comparable to or larger than English and Spanish, each with a bootstrap interval excluding zero. The cross-language correlation is likewise near zero (Spearman’s $\rho = -0.010$), but is underpowered at this sample size and the claim does not rest on it. Subword fragmentation carries information beyond the lexical reference, but with the opposite sign to the standard account: heavier fragmentation predicts dense retrieval performing relatively better, because it damages lexical matching more ($\rho = -0.565$); this association is detected but imprecisely estimated ($\rho = +0.532$, interval $+0.069$ to $+0.806$). We further show that judgment sparsity in this benchmark is confounded with language resource level, and that retrieval-trained multilingual encoders have converged on a single subword vocabulary, both of which constrain how this kind of study can be run and interpreted. All results are reproducible from a public benchmark, fixed seeds, and hashed corpus samples.

I. INTRODUCTION

MULTILINGUAL dense retrievers project a query and documents to a common vector space to measure the relevance without relying on lexical overlap. The behaviour they report is not consistent over languages and a typical summary would be that they do well on high-resource languages and do worse on low-resource languages. When a mechanism is suggested, it is typically tokenization, where multilingual vocabularies tend to split high-resource words into fewer pieces and low-resource ones into many pieces, and the representation of the latter is said to suffer.

Both components of that statement can be tested and they are seldom tested in tandem. It will take care to get right on three aspects which can be done wrong.

Collection size. The number of distracting documents, regardless of language, increases retrieval difficulty. The per-language corpora range from 49,043 passages (Yoruba) to 13,306,000 (French), a ratio of $271\times$. Using raw effectiveness in those collections mixes language with the number of collection pieces.

Evaluation geometry. Judged documents have to be kept in any sampled collection, or the number of judged documents becomes undefined, and that number scales with the number of

evaluated queries. Left uncapped, a large portion of the collection of a language with numerous queries will soon be judged, and retrieval will become easier for reasons other than language.

The reference system. A baseline free of the factor under study is needed to attribute a deficit. A lexical baseline has no pretrained parameters and so cannot inherit a pretraining-volume effect, but only if it is applied uniformly. Selecting “the best available analyzer per language” would import exactly the confound being measured, because analyzer maturity is itself resource-correlated.

This paper reports a pre-registered study built around those three controls. Our contributions are:

1. We examine the relationship between dense-over-lexical advantage and pretraining resource volume for all 18 MIRACL languages, with collection size and evaluation geometry equalised, and find that there is no relationship.
2. We investigate whether that advantage can be accounted for by subword fragmentation, and find that it has the *opposite* sign to the usual account.
3. We provide a quantitative measure of pool bias and show that it is entangled with the variable such studies aim to measure.
4. We show that tokenizer effects in multilingual retrieval cannot currently be isolated observationally, because retrieval-trained multilingual encoders have converged on a single vocabulary.

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KEYWORDS

dense retrieval, evaluation methodology, low-resource languages, multilingual retrieval, tokenization.

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5. We classify failures and show the dense advantage is predominantly recall rescue rather than reranking.

We neither propose a new method, nor a new model, nor a new benchmark, nor state-of-the-art effectiveness.

Contributions (3) to (5) are not comments on this study specifically, but on this type of study in general. Our particular design is not the only one that can have a $271\times$ spread in collection size, a judgment density that tracks the variable under investigation, and a model population that has converged on a single vocabulary. All of these constrain what a multilingual retrieval comparison can establish, and all of them can be measured in advance by anyone who looks.

II. RELATED WORK

A. Multilingual retrieval benchmarks

MIRACL [1] offers human relevance judgments in 18 typologically diverse languages and supplies BM25, mDPR and hybrid baselines for the ad hoc retrieval task. It is our evaluation substrate, and published baselines are used to calibrate our lexical reference.

B. Language-specific retrieval degradation

The closest prior work to ours is the Amharic study of Alemneh, Mekonnen and de Rijke [2], which finds a relative MRR@10 gap of 23% between the best zero-shot multilingual retriever and a monolingual Amharic model, and attributes part of the difficulty to over-segmentation. That study is explicit that its “empirical evidence is limited to Amharic”, that it was not able to experimentally isolate tokenization, and that it treats tokenizer and pretraining distribution as inseparable. We directly address the first limitation by testing 18 languages, and the second as far as observational data permits (Section G).

Goworek *et al.* [3] examine the factors behind cross-lingual ranking, relating retrieval effectiveness to typological similarity between query and document languages. Their setting differs from ours in task (cross-lingual versus monolingual retrieval) and in predictor (pair-relative typological distance versus language-intrinsic tokenization and resource measures). They report that their linguistic analyses “rely on aggregate typological resources and broad similarity metrics” and call for broader low-resource coverage.

C. Tokenization and downstream quality

Rust *et al.* [4] isolate tokenizer quality from pretraining data volume by training monolingual models on equally sized corpora with different tokenizers, and find that a dedicated tokenizer matters roughly as much as data size. Their design is the correct way to establish causality and we cannot run it: it requires pretraining. We therefore report associations, not causes, and Section G shows why the observational alternative is also unavailable. Beyond raw fertility, Single-Token Retention Rate has been proposed as a complementary tokenization measure [5]; we report both.

D. Encoders

As our primary retrieval-trained encoder we use multilingual E5 [6], and as a coverage anchor we use LaBSE [7]; XLM-R [8] supplies both the dominant multilingual vocabulary and, through CC-100, our resource proxy. M3-Embedding [9] is included in our tokenizer survey.

III. EXPERIMENTAL DESIGN

The analysis plan was registered before any effectiveness value was computed. The protocol and its four amendments are released alongside this work, each carrying its own date and a statement of whether any results existed at the time it was made. Amendments are additive: the original text is never rewritten, and each records why it was made. Two of the four were made after results existed, and say so.

A. Collections

For each of the 18 MIRACL languages we build a sub-corpus of exactly $N = 49,043$ passages, equal to the full size of the smallest corpus (Yoruba, which therefore uses its complete collection without sampling). Each sub-corpus contains all judged documents for that language’s evaluated queries, plus a uniform random sample of the remaining passages drawn by reservoir sampling over the streamed corpus under a fixed seed. The resulting document-identifier list is hashed (SHA-256) and released so any rebuild can be verified against the registered sample.

B. Queries

Evaluation queries are limited to 300 per language, randomly sampled without replacement under the same seed; languages with fewer (Yoruba 119, Korean 213) use all available. The cap exists because judged-document count scales with query count: uncapped, Arabic would contribute roughly 29,000 judged documents to a 49,043-passages collection, versus about 1,200 for Yoruba. The realised judged fraction is reported per language as a diagnostic and is discussed in Section A.

C. Systems

The lexical reference is BM25 (Okapi, $k_1=1.5$, $b=0.75$) over character 4-grams within whitespace-delimited units, applied *identically* to every language. Uniformity is the point: per-language analyzers were rejected because analyzer maturity is resource-correlated and would inject the confound under study into the reference system. Because Thai, Chinese and Japanese have no word spaces for a purely orthographic reason, character n -grams were chosen instead of whitespace tokenization, which would otherwise contaminate a reference intended to represent language-intrinsic difficulty. $n=4$ is a conventional default and is not tuned per language; sensitivity to n is reported as a robustness check.

The dense system is multilingual E5 (small), used zero-shot and frozen, with mean pooling, L_2 normalisation, maximum sequence length 256, and exact cosine search. Approximate search is not used: at this collection size it would add error without benefit. LaBSE is used as a coverage anchor.

D. Metrics and statistics

The primary metric is nDCG@10; Recall@100 and MRR@10 are secondary. Per-language effects are computed as paired per-query differences on identical queries and identical collections, with 95% confidence intervals from a bootstrap over queries (10,000 resamples, fixed seed). Effects are aggregated by DerSimonian–Laird random-effects meta-analysis, which weights each language by the inverse variance of its estimate; this allows Yoruba to remain in the analysis on 119 queries without distorting the aggregate. Cross-language associations are reported as Spearman correlations and are **descriptive only**: with $k \leq 18$ languages and correlated predictors they are underpowered, and no causal reading is licensed.

Table 1: Per-language nDCG@10. Dense is multilingual E5; BM25 uses uniform character 4-grams. Yoruba is separated because the encoder does not declare support for it.

Lang	BM25	Dense	Δ	95% CI
zh	0.1922	0.7112	+0.5191	[+0.476, +0.561]
ko	0.4051	0.7406	+0.3355	[+0.286, +0.386]
ja	0.4657	0.7913	+0.3256	[+0.286, +0.364]
fr	0.4096	0.7061	+0.2965	[+0.260, +0.333]
th	0.5410	0.8374	+0.2964	[+0.255, +0.337]
te	0.6412	0.9285	+0.2873	[+0.243, +0.332]
ru	0.5189	0.7991	+0.2803	[+0.242, +0.319]
de	0.4856	0.7460	+0.2604	[+0.221, +0.299]
sw	0.4933	0.7527	+0.2594	[+0.215, +0.304]
fa	0.5056	0.7397	+0.2340	[+0.193, +0.274]
ar	0.6238	0.8547	+0.2309	[+0.198, +0.264]
es	0.5963	0.8218	+0.2255	[+0.193, +0.260]
en	0.5904	0.8067	+0.2163	[+0.182, +0.250]
bn	0.5444	0.7541	+0.2096	[+0.172, +0.247]
id	0.5593	0.7364	+0.1771	[+0.139, +0.216]
hi	0.4887	0.6608	+0.1720	[+0.130, +0.214]
fi	0.6999	0.8465	+0.1466	[+0.111, +0.183]
<i>Encoder does not declare support:</i>				
yo	0.5170	0.4494	-0.0676	[-0.158, +0.026]

IV. RESULTS

A. Dense versus lexical retrieval

Table 1 reports per-language results. Dense retrieval is superior in every language the encoder declares support for. The random-effects pooled difference over those 17 languages is +0.2627 nDCG@10, 95% CI [+0.2240, +0.3013], with $I^2 = 94.0\%$. The heterogeneity is substantial and real: the magnitude of the advantage differs considerably across languages, so the pooled value should not be read alone.

Yoruba is reported separately because the encoder does not declare support for it. It is the only language with a negative difference (-0.0676), and its confidence interval includes zero.

B. The advantage does not track resource level

Using CC-100 corpus volume as the pretraining-resource proxy, which spans 1.1 MB (Yoruba) to 82 GB (English), roughly $76,000\times$, the association between the dense advantage and log resource volume is $\rho = -0.010$ over the 17 supported languages. **This correlation is underpowered and we do not rest the claim on it:** its 95% confidence interval is [-0.488, +0.473], and at $k=17$ a true association below $|\rho| \approx 0.5$ could not be resolved. Reporting it as “no association” would be absence of evidence presented as evidence of absence.

The better-powered argument is the per-language pattern itself. The three lowest-resource languages in the set, Swahili (332 MB), Telugu (536 MB) and Bengali (860 MB), show advantages of +0.2594, +0.2873 and +0.2096 respectively, comparable to or larger than English (+0.2163, 82 GB) and Spanish (+0.2255, 14 GB). Each of those per-language intervals excludes zero. A degradation account predicts the reverse ordering, and does not survive it. The lexical reference likewise shows essentially no association with resource volume ($\rho = -0.059$), which is the behaviour a pretraining-free reference should exhibit and which we verify rather than assume.

Fig. 1 shows why the per-language pattern, rather than the correlation, carries the argument: the lowest-resource languages sit among the *largest* advantages, not the smallest. Table 2 shows the pooled estimate is stable when the languages our uniform reference handles worst are removed, and that the null persists in every variant.

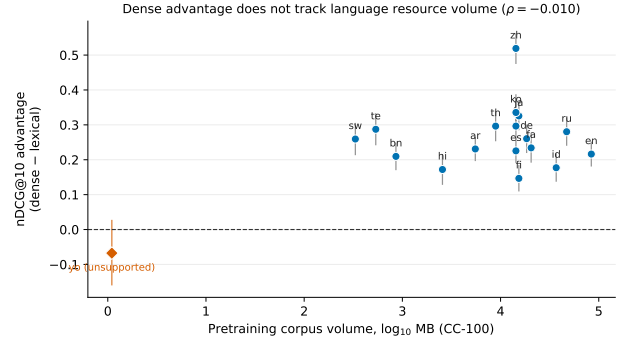


Fig. 1. Per-language nDCG@10 advantage of dense over lexical retrieval against pretraining corpus volume, with 95% bootstrap intervals. Swahili, Telugu and Bengali, the three lowest-resource languages, show advantages comparable to or larger than English and Spanish. Yoruba (diamond) is the one language the encoder does not declare, and is the only point below zero.

Table 2: Sensitivity of the pooled estimate. The pre-registered primary is $k=17$; the exclusions are sensitivity analyses, not a re-selected headline.

Language set	Pooled Δ	I^2	ρ vs. resource
All supported ($k=17$)	+0.2627	94.0%	-0.010
Excluding zh ($k=16$)	+0.2464	86.4%	-0.038
Excluding zh, ko, ja ($k=14$)	+0.2348	82.9%	-0.101

C. Fragmentation, with the opposite sign

Measuring tokenization as subword tokens per character, well defined for every script unlike tokens per whitespace word, which is meaningless for Chinese, Japanese and Thai, the dense advantage is *positively* associated with fragmentation ($\rho = +0.532$, 95% CI [+0.069, +0.806]), and this survives conditioning on the lexical reference (partial $\rho = +0.376$). The interval excludes zero but its lower bound is close to it, so the association is detected rather than precisely estimated. Restricting to segmented scripts weakens but does not remove the association ($\rho = +0.332$, partial +0.292, $k=14$).

The mechanism is visible in the same data: the lexical reference is *negatively* associated with fragmentation ($\rho = -0.565$). Heavier fragmentation damages lexical matching more than it damages dense retrieval, which widens rather than narrows the dense advantage. Collinearity between fragmentation and resource volume is moderate ($\rho = -0.390$), so the two predictors are more separable in this sample than anticipated.

D. Calibration against published baselines

Because 17 of our 18 collections are sampled, absolute values are not comparable to published full-corpus numbers. Yoruba is the exception: its sub-corpus is its complete corpus and it uses all 119 dev queries, so it permits a like-for-like check against the baselines distributed with MIRACL.

Two things follow (Table 3). First, our dense system reproduces the published dense baseline on this language almost exactly (0.4494 versus 0.444), which is a useful check on the pipeline. Second, the reason our comparison shows a dense deficit on Yoruba where the published pair shows a dense *advantage* (0.444 > 0.406) is that *our lexical reference is stronger*, not that our dense system is weaker: 0.5170 against a published 0.406.

This sharpens the interpretation of Yoruba considerably. The

Table 3: Yoruba, the one like-for-like comparison against published MIRACL baselines (nDCG@10). Our dense system lands within 0.006 of the published dense baseline; our lexical reference is considerably stronger than the published one.

System	nDCG@10
Published BM25	0.406
Published mDPR	0.444
This work, dense (E5)	0.4494
This work, lexical ($n=4$)	0.5170
Published BM25 + mDPR hybrid	0.611

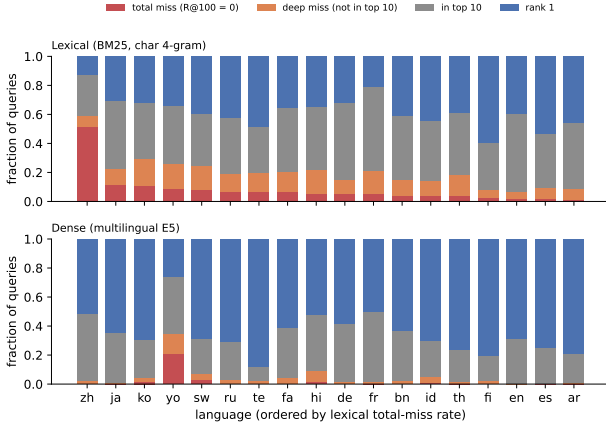


Fig. 2. Failure composition per language, ordered by lexical total-miss rate. The dense system (lower panel) largely eliminates the total-miss category that dominates the lexical reference (upper panel) in Chinese, Japanese and Korean. Yoruba is the exception: dense retrieval introduces total misses rather than removing them.

result is not evidence that dense retrieval is unusually bad in Yoruba (it matches the published dense system there); rather, a uniformly applied character n -gram reference is unusually *good* in Yoruba relative to the analyzer-based baseline. It also reinforces Section D’s wider point: conclusions about a “dense deficit” are sensitive to how strong the lexical comparator is, which is precisely why we fixed ours in advance and report its sensitivity in full.

E. How the systems fail

Categorising each query by whether nothing relevant was retrieved at all (Recall@100 = 0), whether relevant material was found but ranked below ten, or whether it was ranked within ten, the dense advantage is overwhelmingly *recall rescue*: across the 16 supported languages other than Yoruba, dense retrieval rescues 7–26% of queries from a miss category while breaking 0–3%. It nearly eliminates total misses.

Fig. 2 makes the mechanism visible: the dense system converts total and deep misses into ranked hits across almost every language, which mean effectiveness alone conceals.

Two languages behave differently. For Chinese the lexical reference retrieves nothing relevant for 51.3% of queries, so the large apparent dense advantage there reflects reference collapse rather than dense strength. For Yoruba the dense system *introduces* a recall failure: its total-miss rate (0.210) exceeds the lexical reference’s (0.084), and it is the only language where dense breaks more queries (0.227) than it rescues (0.143).

F. Why Yoruba fails

Yoruba is the only language where the dense system underperforms, so we examine its failures directly. Partitioning its 119 queries by outcome, the 27 queries the dense system loses that the lexical reference answers have a mean subword fertility of 2.028 tokens per word, against an all-query mean of 1.951; the 14 queries *both* systems miss have the *lowest* fertility of any group (1.858). **Fragmentation therefore does not explain which Yoruba queries fail**, which is notable because tokenization is the explanation most often offered for such failures.

Qualitatively, the lost queries are predominantly entity-centric factoid questions; proper nouns are identifiable without fluency in the language. This is consistent with lexical matching succeeding on rare named entities where an encoder that does not declare the language has no reliable representation. We state this as an observation rather than an analysis: the author has no Yoruba fluency and no annotator access, so no semantic or morphological judgement of these queries is made.

G. Tokenizer convergence

We surveyed the vocabularies of publicly available multilingual encoders. Every retrieval-trained model examined, namely E5 (small, base, large), BGE-M3, GTE multilingual, Arctic-Embed v2, and Jina v3, uses an XLM-R-derived vocabulary of approximately 250,000 units. The only substantially different vocabulary among the models we tested belongs to LaBSE (501,153), which is trained for bitext mining rather than retrieval and which, in our measurements, loses to the lexical reference on all three languages tested (Yoruba −0.288, Swahili −0.195, Telugu −0.423), including two it fully declares.

Tokenizer and training objective are therefore confounded in the available model population: any pair of models differing in tokenizer also differs in objective. Combined with the impossibility of the pretraining-based design of Rust *et al.* [4] without substantial compute, this means tokenizer effects in multilingual *retrieval* currently cannot be isolated, either experimentally or observationally.

H. Robustness checks

Character 4-grams outperform whitespace tokenization in 17 of 18 languages (mean +0.0758 nDCG@10); Indonesian is the single exception (−0.0542). Chinese under whitespace tokenization scores 0.0071, confirming the absence of word boundaries.

Varying the n -gram order changes the reference substantially, and we report the full sweep rather than a comparison against a single alternative (Table 4). Averaged over languages, $n=3$ is the strongest uniform reference (mean 0.5363, best in 12 of 18 languages), ahead of our registered $n=4$ (0.5154). Our pre-registered choice is therefore *not* the optimum, and we retain it because changing a registered parameter after seeing results is the failure that pre-registration exists to prevent.

What matters is that the conclusion survives the strongest reference we can build. Against $n=3$ the pooled advantage is +0.2403, 95% CI [+0.2082, +0.2724], lower than under $n=4$ (+0.2627) but comfortably positive. The result is not an artefact of a weak baseline. The resource-level null likewise holds throughout ($\rho = +0.167, +0.049$ and -0.010 at $n = 2, 3, 4$), all well inside the wide interval reported above.

Order also interacts with script exactly as expected: $n=2$ is best only for Chinese, Japanese and Korean, languages whose meaningful units are one or two characters, while being

Table 4: Character n -gram sensitivity for the lexical reference (mean nDCG@10 over 18 languages). $n=4$ is the pre-registered primary and is retained despite not being the optimum.

	$n=2$	$n=3$	$n=4$	$n=5$
Mean nDCG@10	0.4219	0.5363	0.5154	0.4811
Best in (of 18)	3	12	3	0
Pooled dense Δ	+0.3548	+0.2403	+0.2627	—
ρ vs. resource	+0.167	+0.049	−0.010	—

markedly worse elsewhere (German 0.4879 at $n=3$ versus 0.2887 at $n=2$). Uniform treatment has a real and now quantified cost.

The Yoruba result is stable across the two strongest references (−0.0659 at $n=3$, −0.0676 at $n=4$) and reverses only under $n=2$ (+0.0553), the weakest reference overall. All three intervals include zero. The defensible statement is therefore that Yoruba is the one language where we cannot demonstrate a dense advantage, and that this does not depend on the registered n except under a baseline configuration that is worse everywhere else. Ablating the E5 query and passage prefixes changes nDCG@10 by −0.0003 on Telugu and +0.0078 on Hindi, the latter chosen in advance as the highest-powered available test because it is the supported language where the encoder scores lowest.

I. Bounding the direction of pool bias

Section A establishes that judgment sparsity is confounded with resource level. That alone does not say which way it pushes our comparison, so we measure the asymmetry directly: how many of each system’s top ten documents are unjudged, per language.

The result is one-sided. In *all* 17 languages the dense system retrieves *fewer* unjudged documents than the lexical reference, by a mean of 1.84 documents out of ten (range −0.33 for English to −4.80 for Chinese). Since an unjudged document is scored as non-relevant by default, the lexical reference absorbs more uncredited retrieval than the dense system does in every language we measure.

The consequence runs against our own result. If some unjudged documents are in fact relevant, which is the premise of the pool-bias concern, then the lexical reference’s true effectiveness is understated by more than the dense system’s, and the advantage we report is therefore an *over-estimate* of the true difference. The association between the effect size and the exposure asymmetry ($\rho = -0.574$) is consistent with that reading: the languages where the dense system’s top ten is most fully judged relative to the lexical one are also the languages where the measured advantage is largest.

We cannot convert this into a numerical correction without relevance judgments we do not have. What it does establish is the *sign*: pool bias in this benchmark inflates rather than deflates the difference we report. Readers should treat the pooled estimate as an upper bound on the true advantage, and this is a stronger caveat than the usual observation that unjudged documents add noise.

J. The conclusion under a second metric

Primary inference above uses nDCG@10. Because this paper argues that measurement choices change conclusions, it would be inconsistent to rest inference on a single cutoff. Repeating the pooled analysis on Recall@100 gives +0.1055, 95% CI [+0.0683, +0.1426] over the same 17 languages. The advantage is smaller, since BM25’s Recall@100 is already high, leaving less headroom, but it remains positive with an interval excluding

zero, and the resource association remains indistinguishable from zero ($\rho = +0.186$, 95% CI [−0.323, +0.612]).

One conclusion *does* change with the metric, and we report it rather than resting on the more convenient one. Under nDCG@10 the Yoruba difference is −0.0676 with an interval including zero, supporting only the statement that no advantage can be demonstrated there. Under Recall@100 it is −0.1310, 95% CI [−0.2157, −0.0504], which does exclude zero: for this encoder, Yoruba shows a measurable recall deficit. That is consistent with the failure analysis, where Yoruba was the one language in which the dense system introduced total misses rather than removing them. As Section K shows, the deficit is a property of this encoder rather than of the language.

K. A second encoder family

All results above use one retrieval-trained encoder. To test whether the effect is a property of that model, we repeated the comparison with BGE-M3, a different retrieval-trained multilingual family, on identical collections and queries (Table 5).

The four languages span the full resource range in the study, a factor of roughly 76,000 from Yoruba to English. The second encoder shows a positive advantage in all four, averaging +0.2321, close to the +0.2627 pooled across 17 languages for the first. We do not compute a resource correlation on four points; the per-language values are the evidence.

The Yoruba row is informative beyond generalisation. Where the first encoder underperformed the lexical reference by 0.0676, the second exceeds it by 0.1904 on the same queries and the same collection. Yoruba is therefore not a language in which dense retrieval cannot work; it is one in which a particular encoder does not.

We stop short of attributing that contrast to declared language coverage. BGE-M3’s model card enumerates no language list, and its base model declares 94 languages without listing Yoruba, so its coverage of Yoruba is undetermined. If it does not in fact declare Yoruba, the result argues against a coverage explanation rather than for one, since an encoder that does not declare the language would still be outperforming the lexical reference there.

V. DISCUSSION

A. On the resource-level null

Our data do not support the claim that dense multilingual retrieval degrades with language resource level. The advantage over a uniformly applied lexical reference is positive in all 17 supported languages and shows no association with pretraining volume. This does not contradict reports of poor absolute effectiveness in low-resource languages (absolute scores do vary), but it does separate two claims that are often merged: that retrieval is *harder* in some languages, and that dense methods are *differentially* disadvantaged there. We find the first and not the second.

One boundary is visible, though it is weaker than it first appears. Yoruba, the single language our encoder does not declare, is the only language where dense retrieval underperforms under the registered reference, and it does so through a recall mechanism rather than a ranking one. The direction is stable across the two strongest reference configurations and reverses only under the weakest one. Since all the relevant intervals include zero, the honest statement is that Yoruba is the one language where we cannot demonstrate a dense advantage, not that we have demonstrated a dense deficit. Whether declared coverage is what matters remains **an open**

Table 5: Generalisation to a second retrieval-trained encoder family (nDCG@10).

Lang	CC-100	BM25	E5	BGE-M3	E5-BM25	BGE-BM25
yo	1.1 MB	0.5170	0.4494	0.7074	-0.0676	+0.1904
sw	332 MB	0.4933	0.7527	0.7778	+0.2594	+0.2845
te	536 MB	0.6412	0.9285	0.9162	+0.2873	+0.2750
en	82 GB	0.5904	0.8067	0.7691	+0.2163	+0.1787

question in our data: testing it requires a retrieval-trained encoder that declares Yoruba, and no encoder available to us is both. LaBSE declares Yoruba but is not retrieval-trained, and its uniform weakness across languages it does cover means it cannot serve as that test.

B. On pool bias

Judgment sparsity is not neutral with respect to the question being asked. Unjudged documents among the top ten range from 2.00 for English to 9.01 for Telugu, and correlate with resource volume at $\rho = -0.660$. Low-resource languages therefore absorb more pool bias, and any cross-language attribution must account for it. Within-language paired comparisons on identical judgment sets, as used here for the primary result, are far less exposed: on Telugu the unjudged counts are nearly identical across systems (9.01 versus 8.66). The threat is to cross-language attribution, which is precisely why our cross-language analyses are reported as descriptive.

VI. LIMITATIONS AND THREATS TO VALIDITY

Associational, not causal. Encoders differ in training data, objective and tokenizer simultaneously. No causal attribution is made anywhere in this paper.

Statistical power. Cross-language analyses have $k \leq 18$ observations. This is a hard limit on what they can show: the 95% interval on the resource correlation spans $[-0.488, +0.473]$, so a true moderate association ($|\rho| \approx 0.4$) would be undetectable. We therefore do not claim that no relationship exists, only that none is detectable here and that the per-language pattern is inconsistent with the direction a degradation account predicts. The unsegmented scripts are also influential: excluding them moves the fragmentation association from $+0.532$ to $+0.332$.

The lexical reference is handicapped in some languages. Uniform treatment costs accuracy where language-specific analysis would help. Korean remains about 10.6% below the published analyzer-based baseline, and the Chinese reference is not merely weak but broken (51.3% total-miss rate). We accept this cost because per-language analyzers would introduce a resource-correlated confound, but the Chinese comparison in particular should not be read as evidence about dense retrieval.

Sampled collections. For 17 of 18 languages the collection is a 49,043-passage sample, so absolute values are not comparable to published full-corpus numbers. This is deliberate, since equal size is what makes cross-language comparison meaningful, but it means our absolute figures should not be placed beside published leaderboard values.

Encoder coverage. The primary results across all 18 languages use one retrieval-trained encoder. A second family (BGE-M3) was tested on four languages spanning the full resource range and reproduced the advantage in all of them (Section K), and LaBSE on three; but a full second sweep of all 18 was not affordable on the available hardware, so generalisation is demonstrated at the range endpoints rather than exhaustively established.

Resource proxy. CC-100 volume proxies the *base model's* pretraining and does not capture the distillation or contrastive

supervision used to build the sentence encoders. A Wikipedia-based proxy for the mBERT-family models was not sourced.

Single hardware environment. All experiments ran on one CPU-only machine. Timings are feasibility figures, not benchmarks.

VII. CONCLUSION

Across 18 languages under equalised collection size and evaluation geometry, a retrieval-trained multilingual encoder outperformed a uniformly applied lexical reference in every language it declares support for, and showed no detectable relationship between the size of that advantage and pretraining resource volume. The strongest form of that evidence is not the cross-language correlation, which is underpowered at $k=17$, but the per-language pattern: the lowest-resource languages show advantages as large as the highest-resource ones. Claims here are scoped to the encoder families tested. Subword fragmentation carried information about the advantage, but in the direction opposite to the usual account, because fragmentation penalises lexical matching more than dense retrieval. Alongside these, we report that judgment sparsity in a standard multilingual benchmark is confounded with language resource level, and that the convergence of retrieval-trained multilingual encoders on a single vocabulary currently prevents tokenizer effects from being isolated observationally. We release the full pipeline, seeds, and hashed collection samples.

CREDiT AUTHORSHIP CONTRIBUTION STATEMENT

Muhammad Munir Khan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization, Project administration.

DATA STATEMENT

The full experimental pipeline, fixed seeds, and SHA-256-hashed corpus sample manifests are released publicly, as described throughout this manuscript and in the accompanying repository, so that the reported results can be independently rebuilt and verified.

DECLARATION OF CONFLICTS OF INTEREST

No conflict of interest exists.

ACKNOWLEDGMENT

An AI-assisted coding and writing tool was used during this work to support software implementation, experimental workflows, and manuscript drafting. All experiments were executed by the author, and all reported results and numerical values were generated from and verified against the

corresponding experimental code and archived artifacts. The author independently reviewed and validated the methodology, results, interpretations, and conclusions and assumes full responsibility for the content of this work.

Funding: This research did not receive funding.

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